

Study of material removal behavior on R-plane of sapphire during ultra-precision machining based on modified slip-fracture model

Suk Bum Kwon^a, Aditya Nagaraj^a, Hae-Sung Yoon^b, Sangkee Min^{a,*}

^a Department of Mechanical Engineering, University of Wisconsin-Madison, 1513 University Avenue, Madison, WI 53706, United States of America

^b School of Aerospace and Mechanical Engineering, Korea Aerospace University, 76 Hanggongdaehak-ro, Deogyang-gu, Goyang-si, Gyeonggi-do 10540, Republic of Korea

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ABSTRACT

In this paper, the modified slip/fracture activation model has been used in order to understand the mechanism of ductile-brittle transition on the R-plane of sapphire during ultra-precision machining by reflecting direction of resultant force. Anisotropic characteristics of crack morphology and ductility of machining depending on cutting direction were explained in detail with modified fracture cleavage and plastic deformation parameters. Through the analysis, it was concluded that crack morphologies were mainly determined by the interaction of multiple fracture systems activated while, critical depth of cut was determined by the dominant plastic deformation parameter. In addition to this, by using proportionality relationship between magnitude of resultant force and depth of cut in the ductile region, an empirical model for critical depth of cut was developed.

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1. Introduction

Single crystal sapphire has good thermal conductivity, optical transmissivity in addition to superior mechanical strength and wear resistance while maintaining tight tolerances and extremely low surface roughness after processing. In particular, the R-plane of sapphire is used in microwave devices and in microelectronic IC applications involving hetero-epitaxial deposition of silicon.¹ Processing single crystal sapphire has been a challenge due to the hard and brittle nature. In order to overcome the brittleness and to prevent crack initiation, nano scale depth of cut is commonly employed during machining. At this small scale, brittle ceramics such as sapphire exhibit a ductile response through plastic deformation rather than crack initiation and propagation.^{2–4} Understanding the fundamental material removal mechanism and being able to predict the ductile-brittle transition in terms of the machining process parameters are important to develop viable machining strategies and improving throughput during the processing of ceramics by ultra-precision machining.

Many researchers have proposed different analytical models to predict the ductile brittle transition during ultra-precision machining or nano-scale material removal operations. Arif et al proposed a predictive model to estimate the uncut chip thickness based on cutting energy. The

ductile-brittle transition occurs when the specific cutting energy for brittle mode cutting exceeds the value for ductile mode cutting.⁵ In the study by Gu et al on predicting the deformation mechanism in sapphire during nanoscale single grit scratch tests using a nano indenter, a modified elastic stress field model was used to predict the activation of cleavage and residual stresses after machining.⁶ In the case of ultra-precision machining with diamond tools, the stress field is much more complicated and hence an alternative approach to determine the ductile-brittle transition is sought. Venkatachalam et al described the ductile brittle transition based on fracture toughness of single crystal brittle materials.⁷ This model was comprehensive in terms of considering tool geometry, material properties and process conditions. Yoon et al investigated the material response of different crystallographic planes of sapphire during ultra-precision machining and quantitatively predicted the average stress when ductile-brittle transition occurs based on the mode I (crack opening under tensile load) stress intensity factor.⁸

The previous literatures provide a good foundation for the research but do not consider the effect of crystal anisotropy in detail. While machining single crystalline ceramics, crystal anisotropy plays an important role in determining the activation of various slip and fracture systems that determine whether the machined surface is ductile or brittle. While considering the effect of crystal anisotropy, it is also important to look at the activation of multiple slip and fracture systems simultaneously based on the cutting direction. Nowak et al explained the anisotropic deformation and fracture of sapphire during nano indentation using effective resolved shear stress and weighted Schmid factor.⁹

* Corresponding author at: Department of Mechanical Engineering, University of Wisconsin-Madison, Room 1039, 1513 University Avenue, Madison, WI 53706, United States of America.

E-mail address: sangkee.min@wisc.edu (S. Min).

Mizumoto et al followed a similar approach and used resolved tensile stress and critical value of fracture energy to predict the activation of cleavage systems during ultra-precision machining of C-plane of sapphire.¹⁰ Maas et al applied the slip/fracture model on M-, R-plane of sapphire during ultra-precision machining and explained the anisotropic crack morphologies and critical depth of cut with supporting analysis using nano indentation.¹¹

Even though many researchers have proposed different energy and stress-based theories, the material response during ultra-precision machining is not completely understood. As advanced ceramics like sapphire are increasingly adopted across various industries, understanding the fundamental mechanisms of deformation during machining of ceramics is necessary to improve the throughput of the machining process.

This paper focuses on modifying the previous slip-fracture model for a more accurate prediction of the material removal behavior on the R-plane of sapphire during orthogonal ultra-precision cutting by reflecting force angle. The R-plane of sapphire was chosen as it shows a large variation in the critical depth of cut in terms of cutting direction and the differences in the parameters affecting the critical depth of cut per cutting direction are expected to be clearly distinguished. An empirical model predicting the critical depth of cut on R-plane during orthogonal ultra-precision machining was developed.

2. Model development

2.1. Introduction to slip/fracture activation model

Single crystal sapphire consists of several functional crystallographic planes; prismatic A-, basal C-, prismatic M-, and rhombohedral R-planes as shown in Fig. 1. Plastic deformation in a single crystal material is activated through various slip systems represented by a crystallographic plane and a preferred slip direction. Similarly, cleavage fracture is activated through specific fracture systems represented by the crystallographic plane and corresponding plane normal.

Mizumoto et al proposed a plastic deformation parameter and cleavage fracture parameter to predict the activation of specific slip and

fracture systems based on the cutting direction to account for the anisotropy of the crystal as shown below¹⁰:

$$P_i = m_i / (\tau_i^{\text{crit}} / \min_i \tau_i^{\text{crit}}) \quad (1)$$

$$F_j = c_j / (E_j^{\text{frac}} / \min_j E_j^{\text{crit}}) \quad (2)$$

where P_i is the plastic deformation parameter for the i -th slip or twinning system, τ_i^{crit} is the critical resolved shear stress for the i -th slip or twinning system, F_j is the fracture cleavage parameter for the j -th fracture system, and E_j^{frac} is the fracture energy for the j -th fracture system. Both plastic deformation parameter and fracture cleavage parameter have dimensionless unit while unit of critical resolved shear stress and fracture energy are MPa and J/m² respectively. m_i is the Schmid factor and c_j is the cleavage factor synonymous to the Schmid factor as given below:

$$m_i = \cos \theta_i \cos \lambda_i \quad (3)$$

$$c_j = \cos^2 \psi_j \quad (4)$$

where θ_i is the angle between the direction of applied stress and normal to the slip plane, λ_i is the angle between direction of applied stress and slip direction, and ψ_j is the angle between the direction of applied stress and normal to the cleavage plane as shown in Fig. 2. The unit of both Schmid factor and cleavage factor are dimensionless. The activation of slip systems is governed by critical resolved shear stress while activation of cleavage systems is governed by fracture energy listed in Tables 1 and 2, respectively. Eq. (1) indicates the likelihood of activation of a given slip or twinning system. In a similar way, Eq. (2) is used to express likelihood of activation of fracture systems.

2.2. Modified slip/fracture activation model reflecting measured force direction

Fig. 2 shows how the applied force, called resultant force (F_r), contributes to ductile deformation through slip and brittle fracture through

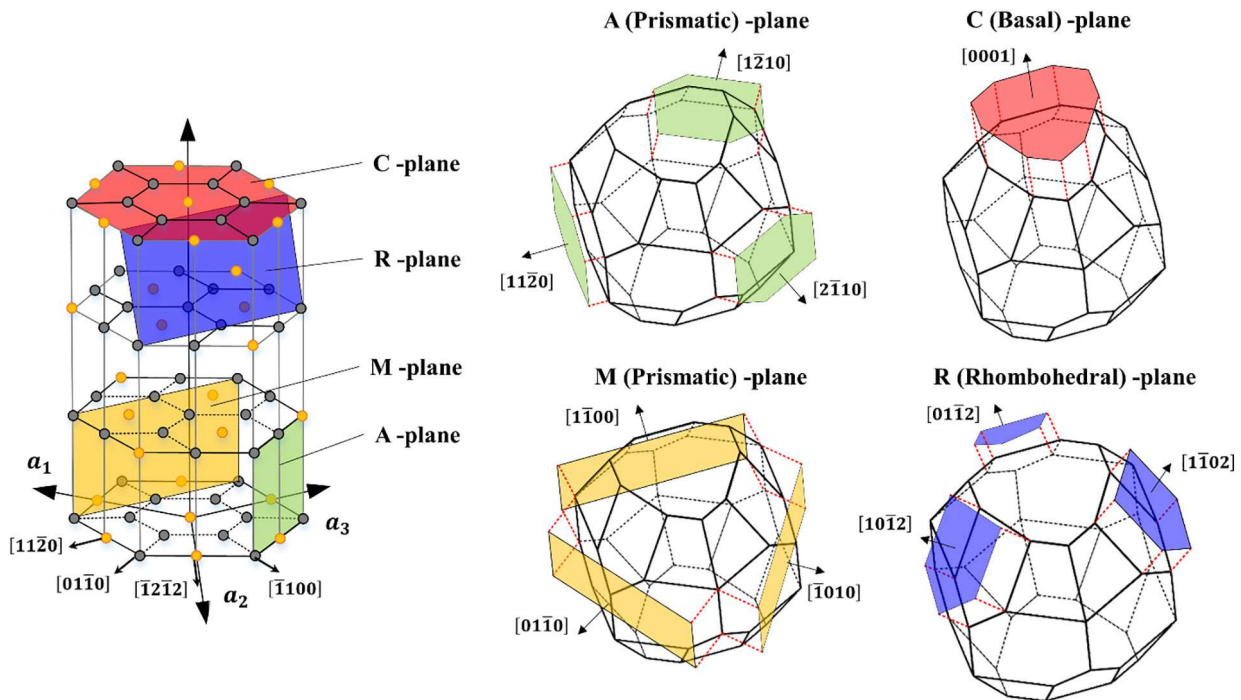


Fig. 1. Functional crystallographic planes of single-crystal sapphire.

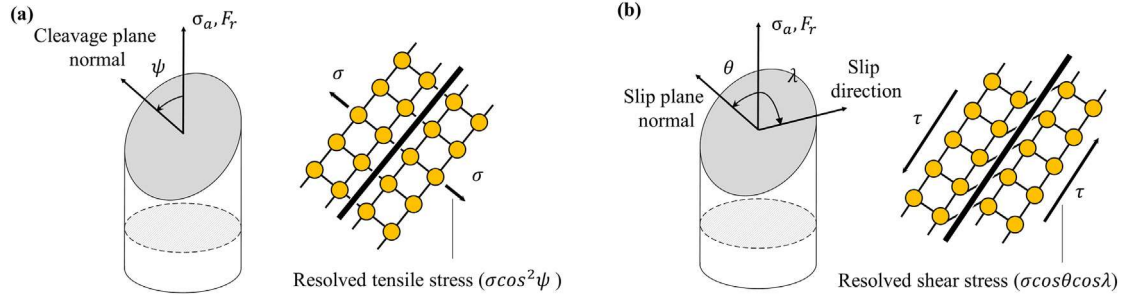


Fig. 2. Schmid factor and cleavage factor.

Table 1
Slip/twinning systems of the single-crystal sapphire.¹²

Slip (twinning) system	Miller-Bravais indices	Critical resolved shear stress (MPa)
Basal twinning (C-plane)	(0001)($\bar{1}010$)	2.2255
Basal slip (C-plane)	(0001)($11\bar{2}0$)	2.2255
Pyramidal slip	($0\bar{1}11$)($10\bar{1}1$)	4.4817
Prismatic slip (A-plane)	($11\bar{2}0$)($\bar{1}100$)	1.6487
Rhombohedral twinning (R-plane)	($\bar{1}102$)($10\bar{1}1$)	0.4066

Table 2
Fracture systems of single-crystal sapphire.¹²

Fracture system	Fracture energy (J/m ²)
Basal fracture (C-plane) (0001)	>40
Prismatic fracture (M-plane) ($10\bar{1}0$)	7.3
Rhombohedral fracture (R-plane) ($\bar{1}012$)	6

crack initiation in terms of Schmid factor and cleavage factor. As shown, the values of θ , λ , and ψ are functions of direction of F_r , as it determines the direction of applied stress and relative angle to plane normal and activation direction.

The direction of resultant force is determined by cutting direction and force angle, θ_{force} as shown in Fig. 3, and the force angle is given by Eq. (5)

$$\theta_{\text{force}} = \text{atan} F_t / F_c \quad (5)$$

where F_t is the thrust force, and F_c is the cutting force. Both F_t and F_c are in units of N, and the unit of θ_{force} is radian. Compared to previous slip/fracture model which assumed force angle as a fixed value, the modified model considers variation of force angle. Based on the Eq. (5), a plastic deformation parameter and fracture cleavage parameter were modified as shown below:

$$P_i' = m_i' / (\tau_i^{\text{crit}} / \min_i \tau_i^{\text{crit}}) \quad (6)$$

$$F_j' = c_j' / K_{ICj} / \min_j K_{ICj} \quad (7)$$

where P_i' and F_j' are the newly calculated plastic deformation parameter and fracture cleavage parameter, K_{ICj} is the mode I critical stress intensity factor for the j -th fracture system with unit of MPa m^{-0.5}, and m_i' is the modified Schmid factor for the i -th slip system reflecting force angle, c_j' is the modified cleavage factor for the j -th fracture system reflecting force angle. And for the modified model, activation of cleavage systems is governed by mode I critical stress intensity factor listed in Table 3 as it represents severity of the stress state generating crack opening under tensile load. The modified plastic deformation parameter and fracture cleavage parameter have dimensionless unit.

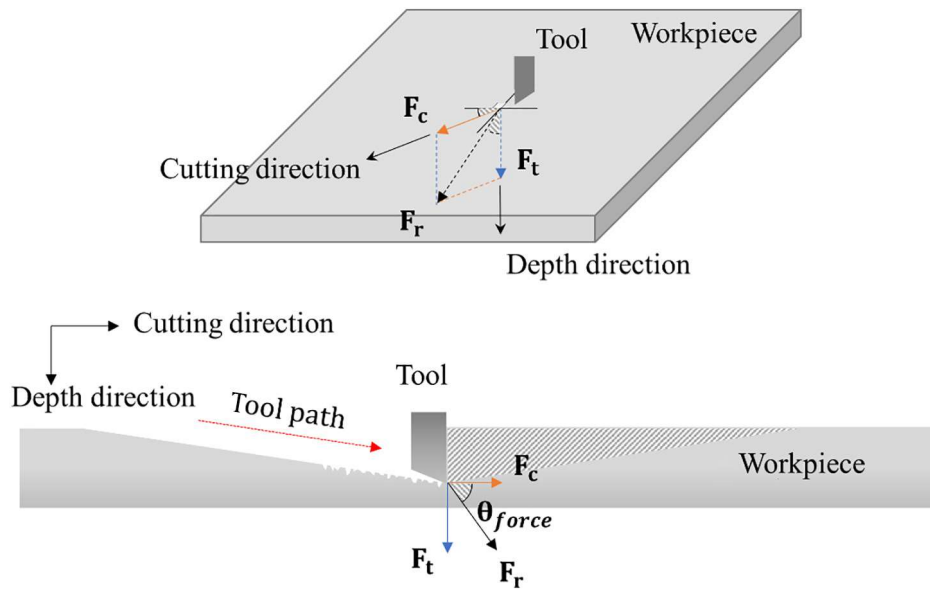


Fig. 3. Direction of resultant force.

Table 3
Fracture systems of single-crystal sapphire.¹³

Fracture system	Critical stress intensity factor (MPa · m ^{-1/2})
Basal fracture (C-plane) (0001)	4.54
Prismatic fracture (M-plane) {10 $\bar{1}$ 0}	3.14
Rhombohedral fracture (R-plane) { $\bar{1}$ 012}	2.38
Prismatic fracture (A-plane) {11 $\bar{2}$ 0}	2.43

2.3. Crack morphology

Wang et al showed that the observed crack morphology after nano-indentation test is dependent on the simultaneous activation of different fracture systems and their intersection at the surface of the material.¹⁴ In this study, this approach is extended to the case of ultra-precision machining and it is believed that the cleavage produced by activation of different fracture systems will propagate until it meets the surface, forming an intersection line on the machined surface resulting in a characteristic crack morphology. Based on the geometrical information of fracture systems, the possible intersection lines that can appear on the machined plane and corresponding angles were calculated as shown in Fig. 4 and Table 4. The labels in Fig. 4 are set according to the type of cleavage plane and its activation. R, B, PRS (A), PRS (M) represent the rhombohedral (R-), Basal (C-), 1st prismatic (A-),

Table 4
Possible fracture cleavage and corresponding angles.

Crack type	Corresponding angle
R_1R_2	47°
R_1R_3	133°
R_1A_1	47°
R_1A_2	133°
R_1A_3	90°
R_1C	180°
R_1M_1	73°
R_1M_2	117°
R_1M_3	180°

and 2nd prismatic (M-) planes, respectively. The machined plane is considered to be R_1 , rhombohedral plane. The subscript is used to denote other similar planes but at different orientations.

3. Experimental setup

For the experiment, plunge cuts were performed on an R-plane sapphire substrate having dimensions of 10 mm × 10 mm × 0.5 mm (Czochralski technique, polished, M.T.I Corp., U.S.A) using a 5 axis ultra-precision CNC machine with 1 nm command resolution in the 3 linear axes and 1 micro-degree command resolution in the 2 rotary

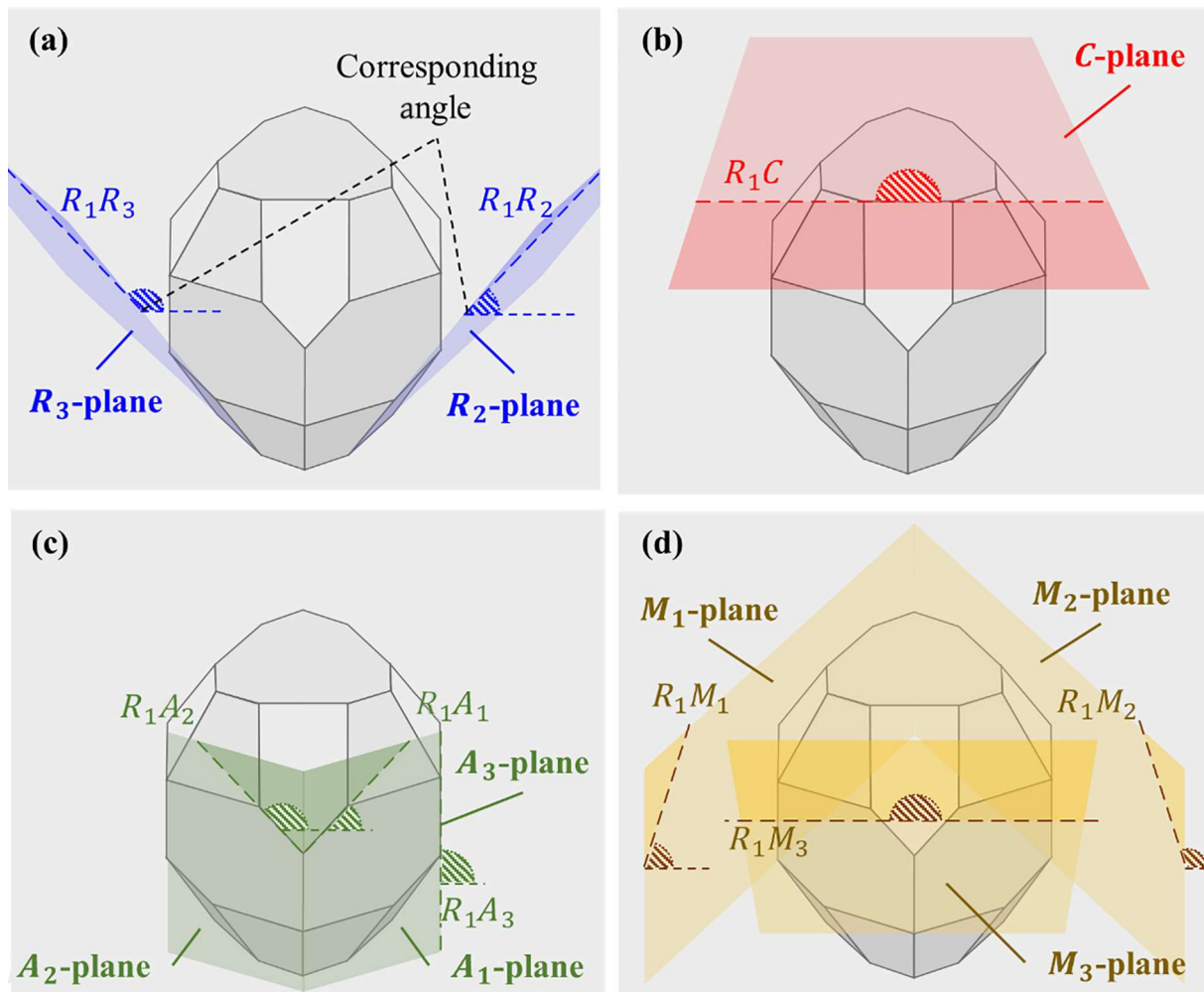


Fig. 4. Crystal structure of single-crystal sapphire and intersection line between R-plane and other fracture crystal planes.

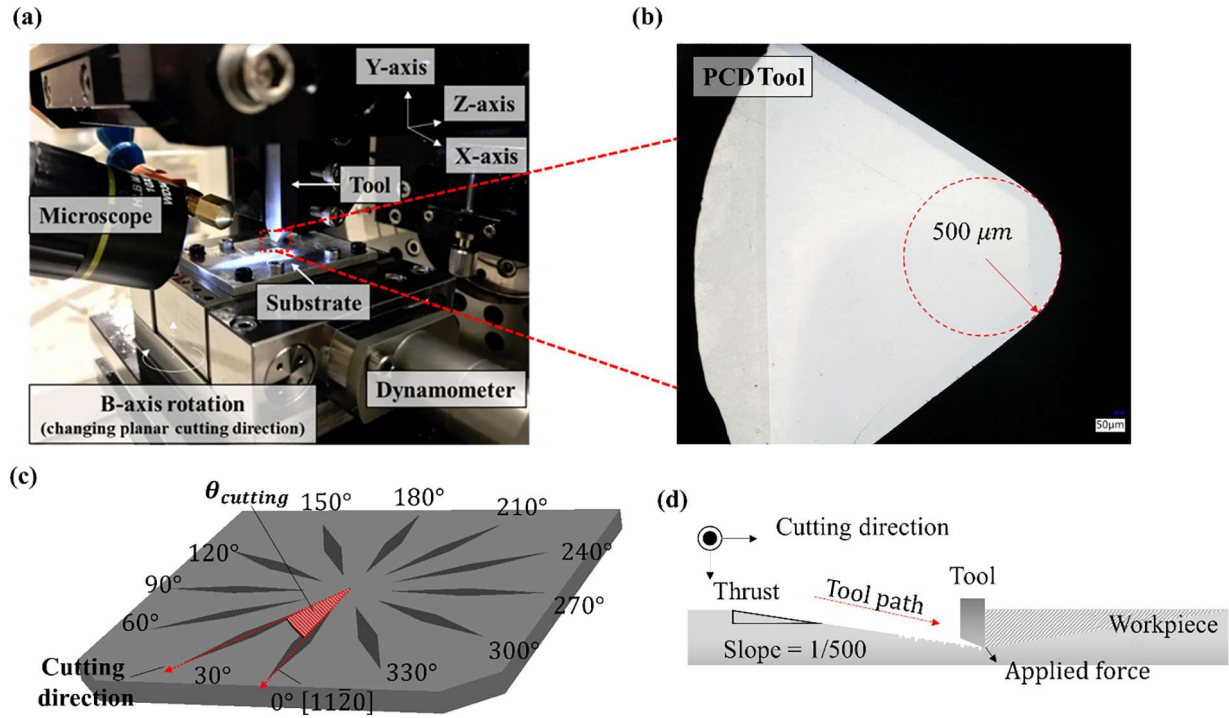


Fig. 5. (a) Experimental setup for ultra-precision cutting on R-plane, (b) top view of the polycrystalline diamond tool, (c) orientation of R-plane workpiece, and (d) schematic view of plunge cutting.

axes, ROBONANO α -0iB (FANUC Corporation, Japan) under continuous supply of mineral oil based coolant. This experimental setup is shown in Fig. 5(a). A binder-less nano-polycrystalline diamond tool with nose radius of 0.5 mm, rake angle of 0°, edge radius of 500 nm, and clearance angle of 9° (A.L.M.T. Corp., Japan) was used to carry out the machining as shown in Fig. 5(b). The specimen was mounted on a piezoelectric dynamometer (Type 9119AA1, KISTLER Instrument Corp., Switzerland) connected to a multi-channel amplifier with low pass filter (Type 5080A, KISTLER Instrument Corp., Switzerland) to measure the cutting and thrust forces. A tilt stage and surface probe were used to ensure flatness of the specimen during the experiments. The plunge cuts were performed in different planar cutting directions at 30° intervals as shown in Fig. 5(c) and with an entry and exit slope of 1/500 at a feed rate of 5 mm min⁻¹ as shown in Fig. 5(d). Each set of experiments was repeated at least 3 times to ensure consistency of the results. Tool-work contact was established using an optical microscope. The machined surfaces were examined using an optical microscope and scanning electron microscope (SEM, Zeiss 1530) to observe the crack morphology and determine the depth at which ductile-brittle transition occurred.

Compared to previous research which assumed the force angle as constant value,^{10,11} the direction of the resultant force in terms of depth of cut was calculated based on measured force profile by dynamometer during the machining. Fig. 6(a) shows the profile of thrust force and cutting force during orthogonal plunge cutting. By applying curve fitting, average cutting force and thrust force were obtained as function of time. The cutting time was converted into depth of cut with information on feed rate and tool path. The force angle over depth of cut was calculated as shown in Fig. 6(b).

4. Result and discussion

4.1. Machining characteristics in terms of cutting directions

Fig. 7 shows results of the crack morphologies and measured critical depth of cut in terms of cutting direction on R-plane during plunge cutting. The critical depth of cut is the depth at which the ductile-brittle transition occurs and is measured by observing the machined surface

through an optical microscope with the order of 500× magnification. It was found that crack morphology and critical depth of cut show a clear difference depending on the cutting direction, since the crystal structure has a significant role in determining the machining behavior.

The variation in critical depth of cut shows symmetry about the line with direction of [0001] (red dotted line), and large differences in numerical value based on the cutting direction. For 0° and 180° the CDCs are approximately 600–700 nm which is relatively high compared to that of other directions, in which the CDC is in the range of 50–200 nm. Fig. 8 shows the SEM images of the brittle regions on the machined slots at a depth of cut 100–200 nm deeper than the critical depth of cut. As the R-plane exhibits two-fold symmetry about [0001] in terms of critical depth of cut and crack morphologies, only cutting directions from 90° to 270° in intervals of 30° were used in the analysis. The morphology of the crack for each cutting direction was categorized into four types as shown in Fig. 8: Type 1 – pull out crack, Type 2 – triangular crack, Type 3 – radial crack, Type 4 – fan-shaped crack. Type 2 and Type 4 cracks were further categorized into several sub-groups based on the severity of horizontal cleavage or fan-shaped cracks with spalling.

4.2. Crack morphology and fracture-cleavage parameter

Fig. 9 shows SEM observations of Type 1 cracks at about 150 nm depth of cut and variation in related fracture-cleavage parameters with the depth of cut. The shape of crack is characterized by a pull-out crack that forms horizontal intersection lines with the machined surface by basal R_1C and prismatic R_1M_2 fracture cleavages planes. In Fig. 10, as the direction of basal R_1C and prismatic R_1M_2 fracture cleavages planes are opposite to one another, simultaneous activation of these two fracture systems meet each other at some depth below the surface and fractured pieces of material are pulled out.

Figs. 11 and 12 show Type 2 cracks in the 120° cutting direction (Type 2-1) around 400 nm depth of cut and in the 150° (Type 2-2) cutting direction around 400 nm depth of cut, with related fracture-cleavage parameters. The noticeable characteristic of Type 2 cracks is the horizontal intersection line of basal R_1C fracture (180°). For both

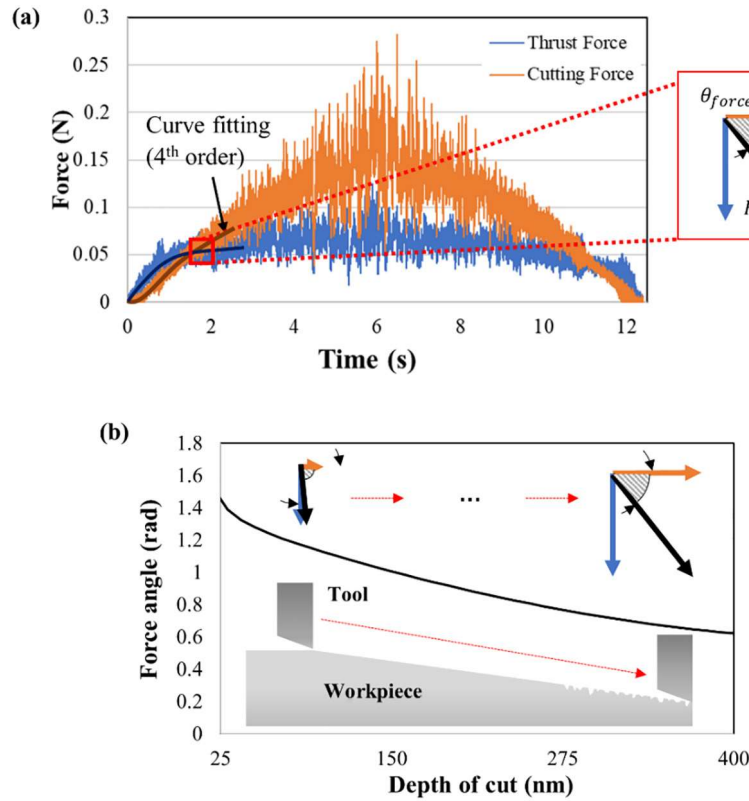


Fig. 6. Calculation of direction of force from measured force profile.

Type 2-1 and 2-2, intersection lines between rhombohedral R_1R_3 fracture (133°), prismatic R_1A_1 (47°) and prismatic R_1M_2 (117°) fractures are visible at the end of the surface created by basal fracture cleavage. In the 120° cutting direction, the horizontal lines formed by R_1C and

R_1R_3 fractures were seen more prominently. This is in line with the prediction for the fracture-cleavage parameters as the R_1R_3 and R_1C cleavage systems have a higher likelihood of activation compared to other fracture cleavage systems, Fig. 11(a). Meanwhile, for the 150° cutting

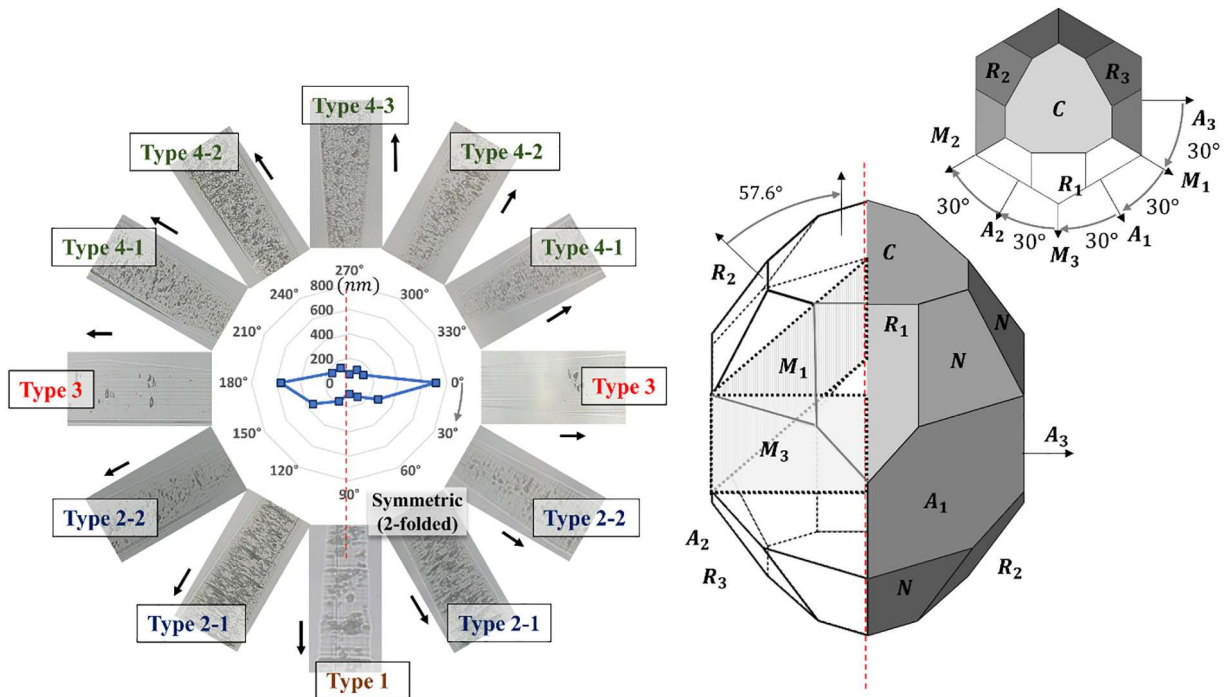


Fig. 7. Critical depth of cut and classification of crack morphology on R-plane in terms of different cutting directions.

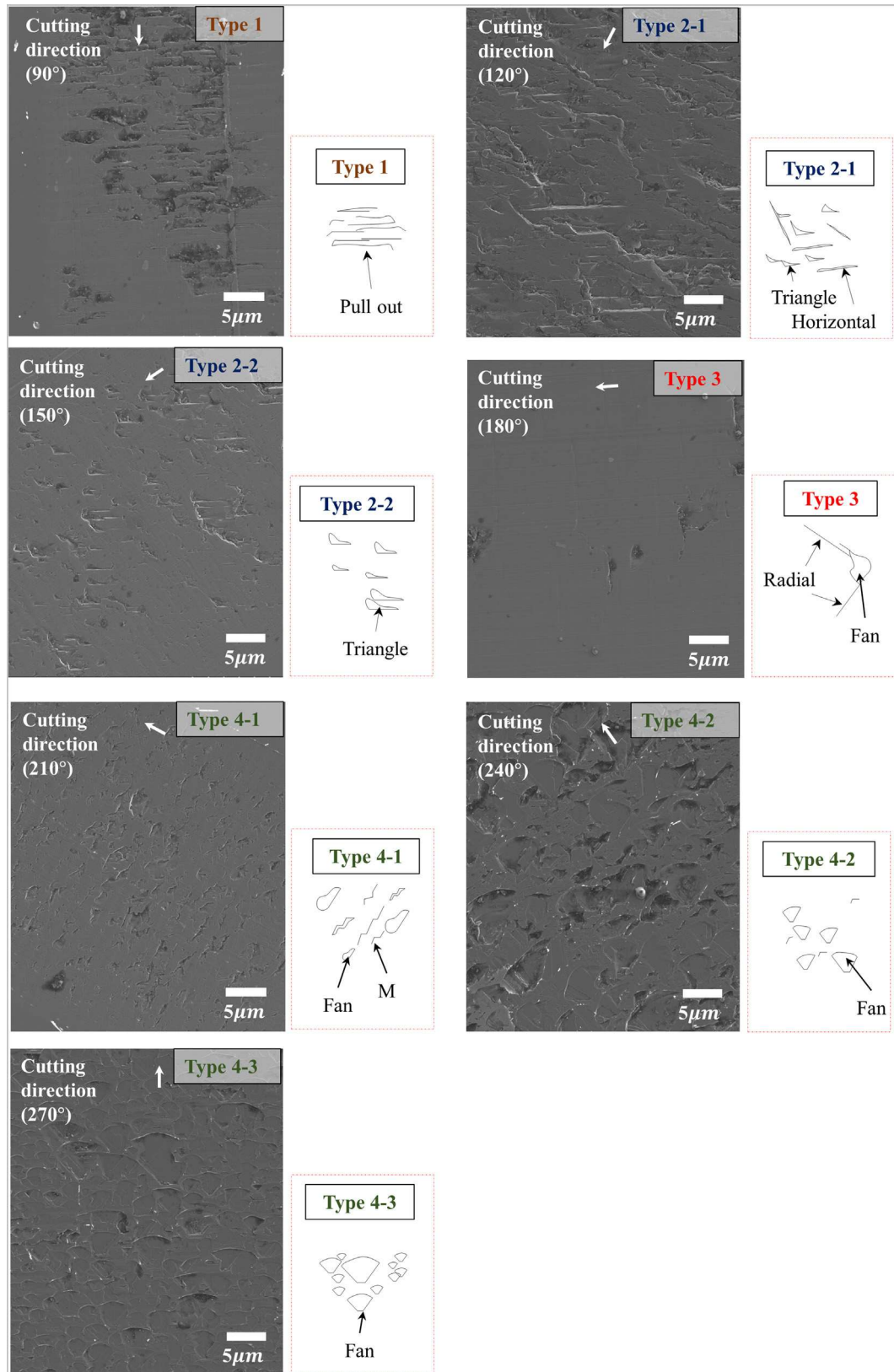


Fig. 8. SEM observation and categorization of cracks on machined surface in terms of different cutting directions.

direction, multiple intersection lines of crack morphology were visible on the machined surface as many of the fracture cleavage parameter have relatively similar values as shown in Fig. 12(a).

Fig. 13 shows the Type 3 cracks in the 180° cutting direction and related fracture-cleavage parameters at 600 nm depth of cut. In Fig. 13(a), the intersection lines of rhombohedral R_1R_2 fracture (47°), R_1R_3 fracture

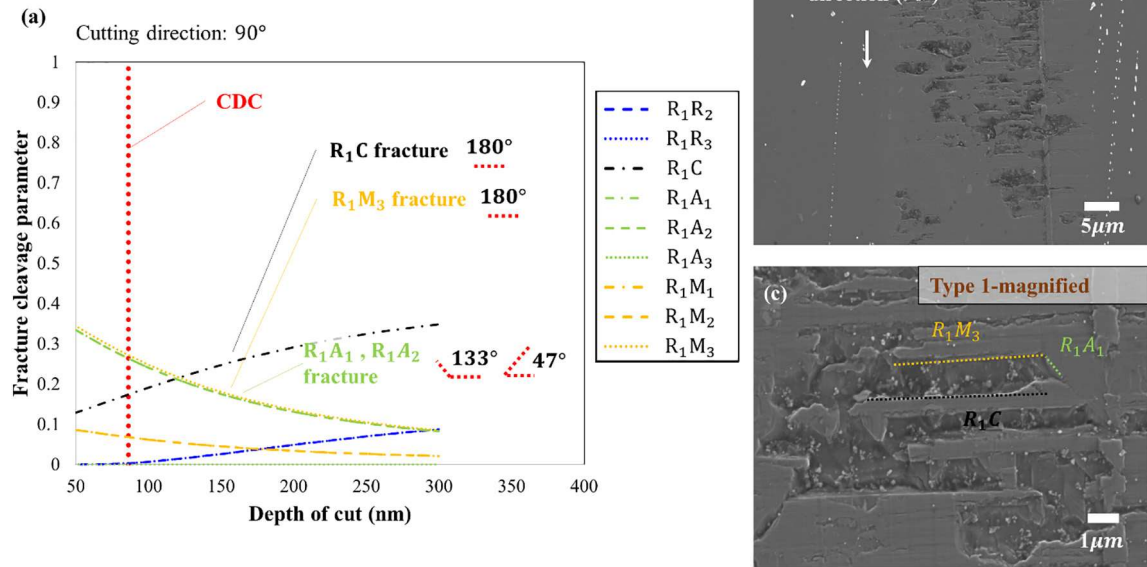


Fig. 9. Crack morphology and fracture-cleavage parameter for Type 1 Crack (90°).

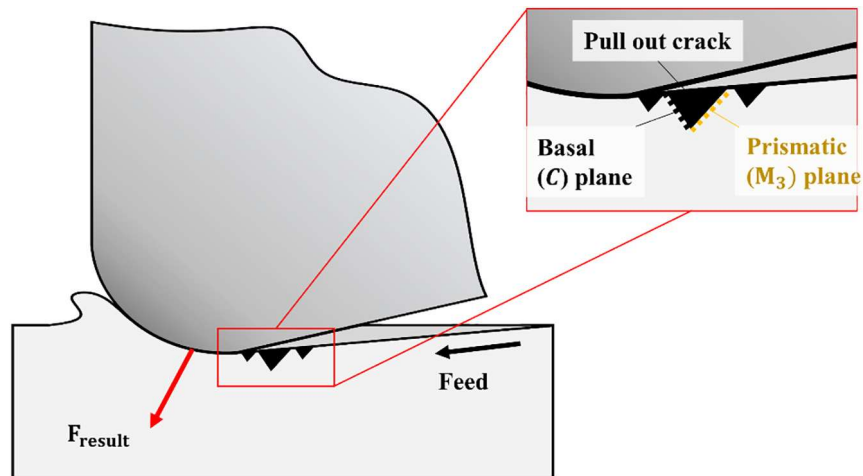


Fig. 10. Pull out crack morphology by prismatic M and basal C fracture cleavages in 90° cutting direction.

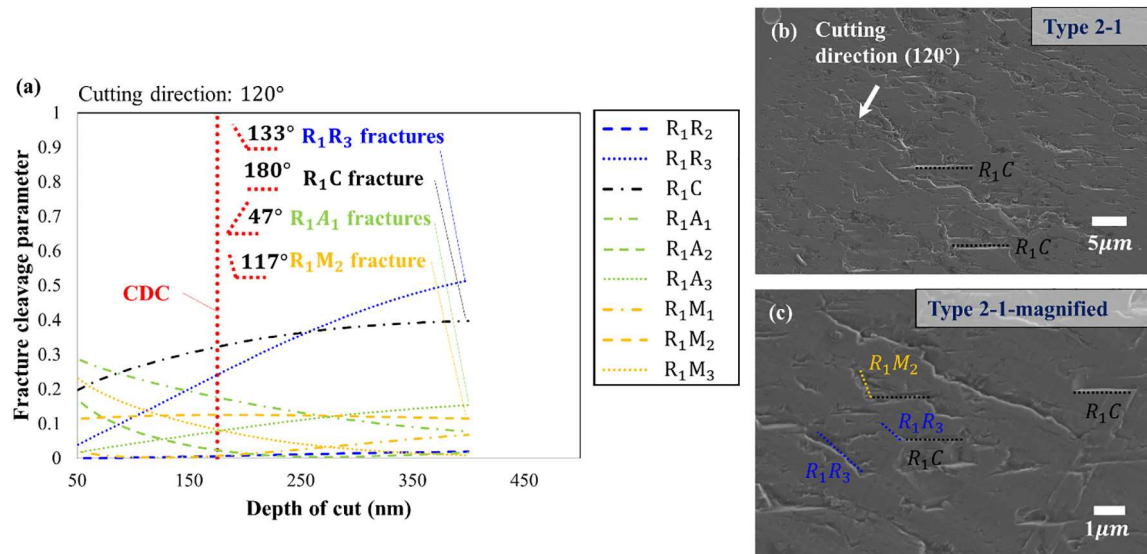


Fig. 11. Crack morphology and fracture-cleavage parameter for Type 2-1 crack (120°).

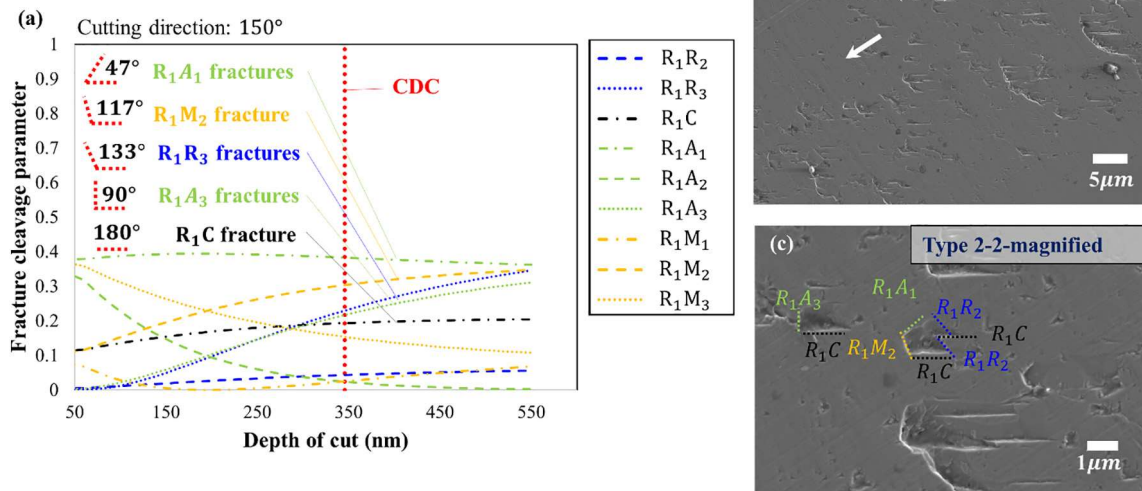


Fig. 12. Crack morphology and fracture-cleavage parameter for Type 2-2 crack (150°).

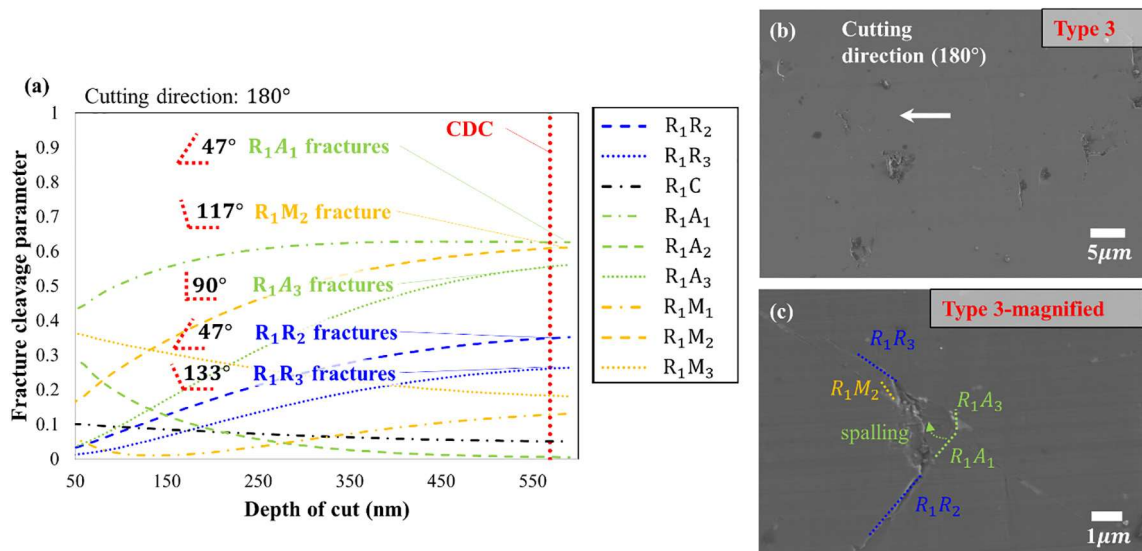


Fig. 13. Crack morphology and fracture-cleavage parameter for Type 3 crack (180°).

(133°), prismatic R_1A_1 (47°), R_1A_3 (90°) and prismatic R_1M_2 (117°) fractures are clearly shown. Fig. 13(c) shows spalling crack and the angle of the boundary lines of bottom is 47°. At both ends of the fan-shaped crack, radial cracks at angles of 47°, 117°, and perpendicular to the

cutting direction were observed. The spalling crack morphology at 180° cutting direction may be related to the high value of the prismatic R_1A_1 fracture-cleavage parameter. Fig. 14 shows the activation of prismatic R_1A_1 fracture cleavage and plastic deformation. As this fracturing

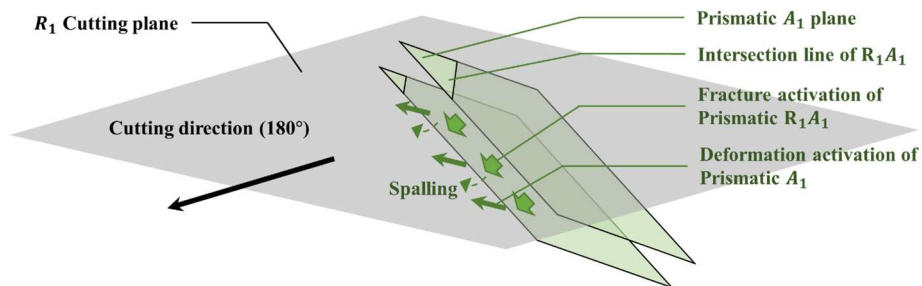


Fig. 14. Spalling crack by prismatic A plane in 180° cutting direction.

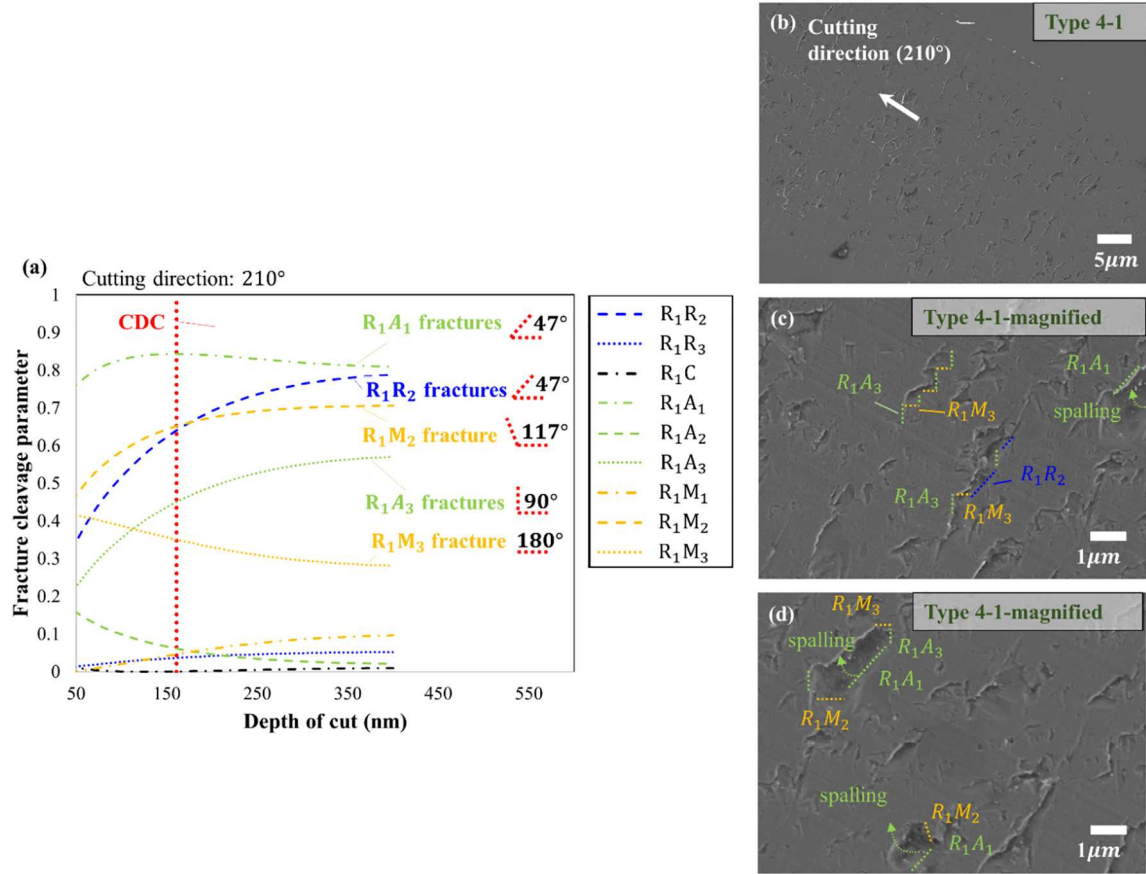


Fig. 15. Crack morphology and fracture-cleavage parameter for Type 4-1 crack (210°).

plane becomes a form of shallow digging with a coupling of prismatic deformation, it appears to make a spalling type crack as a result. Also, the prismatic plastic deformation on the same plane may positively accelerate this fracture activation.

Figs. 15–17 show the Type 4 cracks in 210° (Type 4-1), 240° (Type 4-2), and 270° (Type 4-3) cutting directions and related fracture-

cleavage parameters at 350–400 nm depth of cut. Type 4 cracks are represented by fan-shaped cracks caused by both slip/fracture activation of prismatic R_1A_1 plane. In the 210° cutting direction, M-shaped cracks were dominant and seem to be generated by a combination of rhombohedral R_1R_2 , prismatic R_1A_3 , and R_1M_3 fracture cleavage activations, Fig. 16.

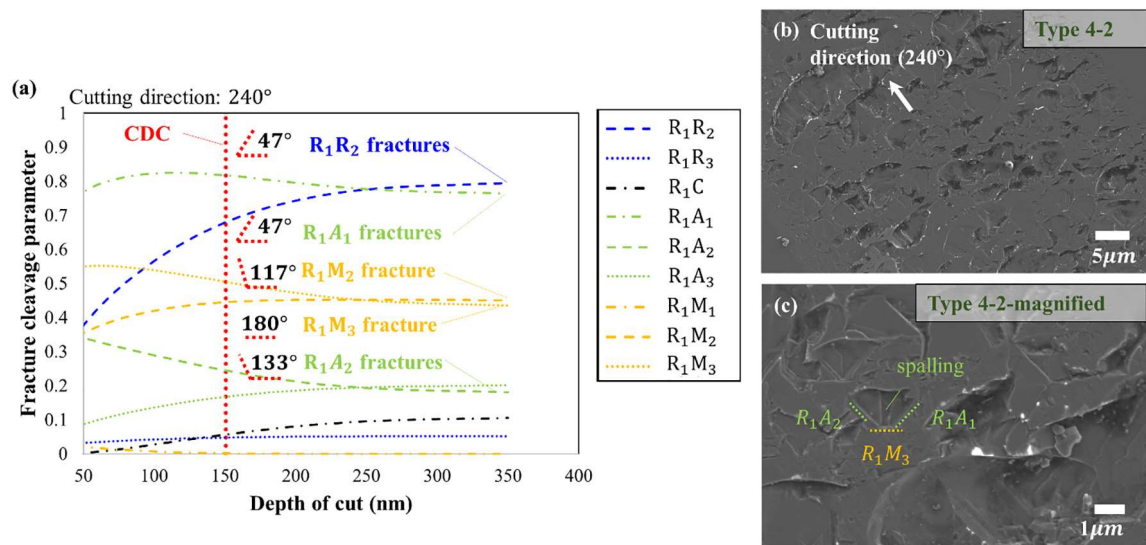


Fig. 16. Crack morphology and fracture-cleavage parameter for Type 4-2 crack (240°).

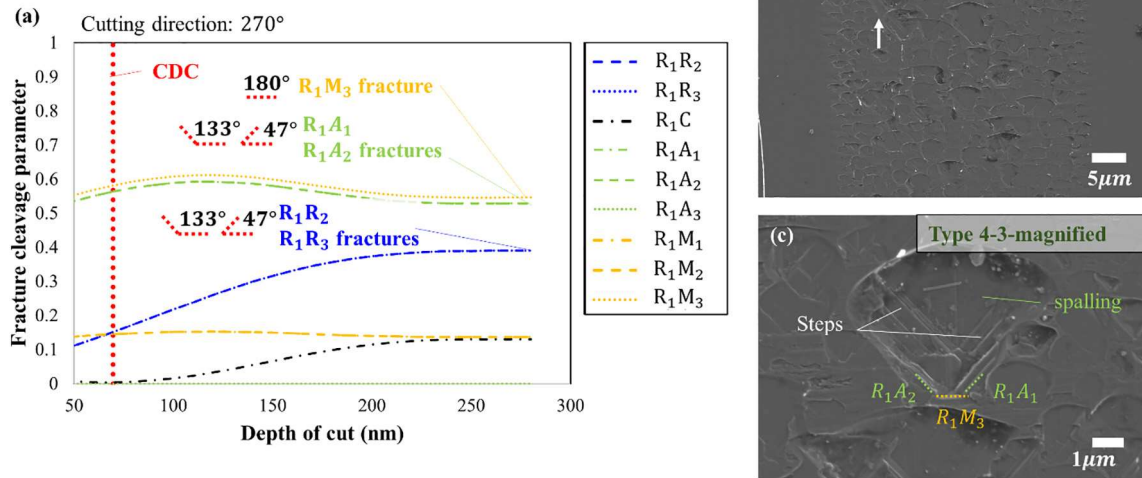


Fig. 17. Crack morphology and fracture-cleavage parameter for Type 4-3 crack (270°).

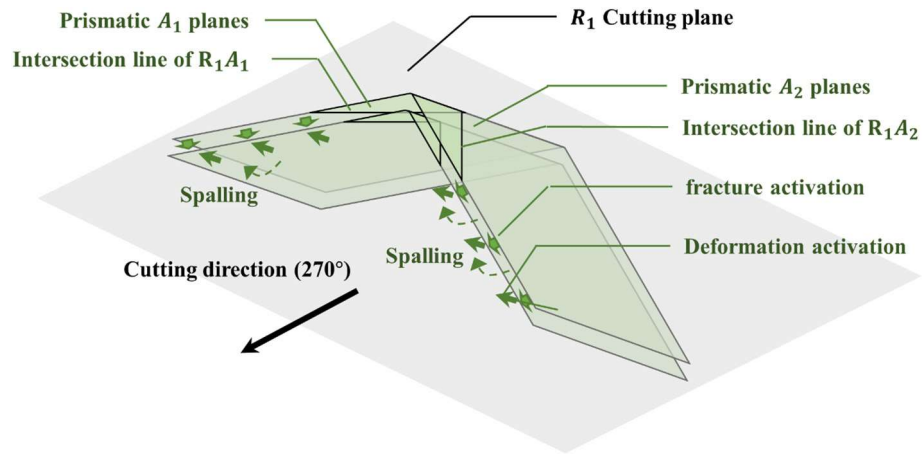


Fig. 18. Spalling crack by prismatic A plane in 180° cutting direction.

Table 5

Crack morphologies and mechanism in terms of cutting directions.

Cutting direction (°)	Crack category	Crack type	Mainly activated fracture system	Mechanism
90	Type 1	Pull out	Basal C- Prismatic M-	Two fracture systems (C-, M-) with opposite activation directions meet and form pull out crack morphology (Fig. 10).
60, 120	Type 2-1	Triangular	Basal C- Rhombohedral R- Prismatic A- Prismatic M-	Horizontal crack morphology by C-fracture activation and diagonal crack morphology by R-, A- and M-fracture activations.
30, 150	Type 2-2	Triangular	Basal C- Rhombohedral R- Prismatic A- Prismatic M-	Horizontal crack morphology (C-) and hybrid crack morphology by R-, A-, M-fractures activations which have a similar fracture activation tendency.
0, 180	Type 3	Radial	Rhombohedral R- Prismatic A- Prismatic M-	Radial crack morphology by R-, M-fracture activations. Fan-shaped spalling crack by A-fracture activation (Fig. 14).
210, 300	Type 4-1	Fan (spalling)	Rhombohedral R- Prismatic A- Prismatic M-	Fan-shaped spalling crack morphology (A-) and M-shaped crack morphology by combination of A- and M-fracture activations.
240, 330	Type 4-2	Fan (spalling)	Prismatic A- Prismatic M-	Fan-shaped spalling crack morphology (A-).
270	Type 4-3	Fan (spalling)	Prismatic A- Prismatic M-	Fan-shaped spalling crack morphology symmetrical to the [0001] direction (activation of symmetric A-fractures at the same time) (Fig. 18).

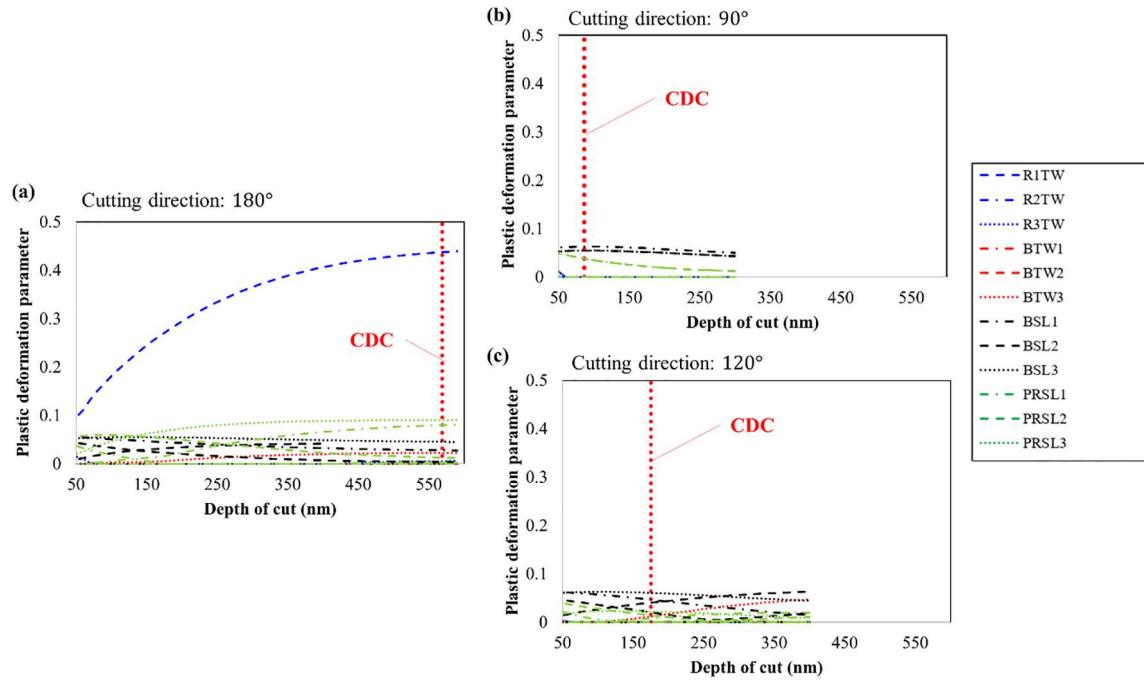


Fig. 19. Comparison of plastic deformation parameter and critical depth of cut in 90° and 120° cutting directions with those in 180° cutting direction.

Cracks due to the activation of both prismatic R_1A_1 and prismatic R_1A_2 planes that are symmetrical to each other. For the cutting directions of 240° and 270°, the fracture-cleavage parameter of prismatic A_1 planes compared to 210° have higher values, Figs. 16 and 17. Hence, more symmetric spalling cracks appeared as the value of prismatic A_2 fracture-cleavage parameter is similar to that of prismatic A_1 fracture-cleavage parameter. Further, fan-shaped cracks with the arc of the fan aligned in the cutting direction are more likely to occur in these cutting directions.

In the 270° cutting direction, the steps found at the diagonal boundary of the fan shaped spalling crack as shown in Fig. 17(c) and it can be explained in the same context as Fig. 14. Fig. 18 shows both fracture cleavage and plastic deformation activation of two symmetric prismatic A planes. With simultaneous activation of prismatic fractures and plastic deformations, symmetric fan-shaped spalling crack is generated with steps with forwarded direction of 47° and 133° can be observed in Fig. 17.

Through the analysis of fracture cleavage parameters, anisotropy in crack morphology as per different cutting directions has been summarized in Table 5.

4.3. Critical depth of cut and plastic deformation parameter

Figs. 19–21 show the critical depth of cut and plastic deformation parameters in terms of cutting directions and depth of cut. The 180° cutting direction shows the largest critical depth of cut and dominant rhombohedral R_2 twinning parameter. In Fig. 19, it can be seen that the 90° and 120° cutting directions have very small plastic deformation parameter values across all slip systems, which results in a low CDC at these angles as shown in Fig. 7. Fig. 20 shows the plastic deformation parameters of 150° cutting direction and shows a trend very similar to 180° cutting direction. For this cutting direction, the critical depth of cut also has a high value (~400 nm). Fig. 21 shows the plastic deformation parameters of 210°, 240°, and 270°. Compared to the case of 180°, there is not as clear a demarcation between the different plastic deformation parameters for these cutting directions which implies the possibility of multiple slip/twinning systems activating simultaneously during cutting. As a result, the activated slip/twinning systems may interact negatively with each other (dislocation entanglement) thus lowering the critical depth of cut.

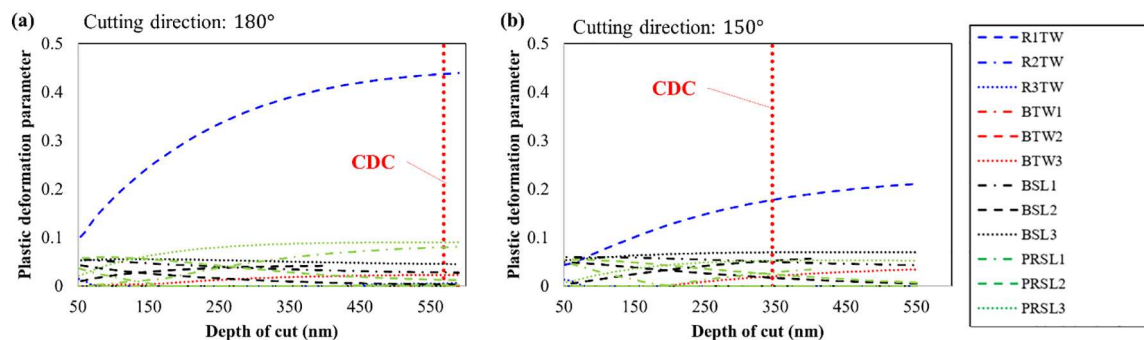


Fig. 20. Comparison of plastic deformation parameter and critical depth of cut in 150° cutting direction with those in 180° cutting direction.

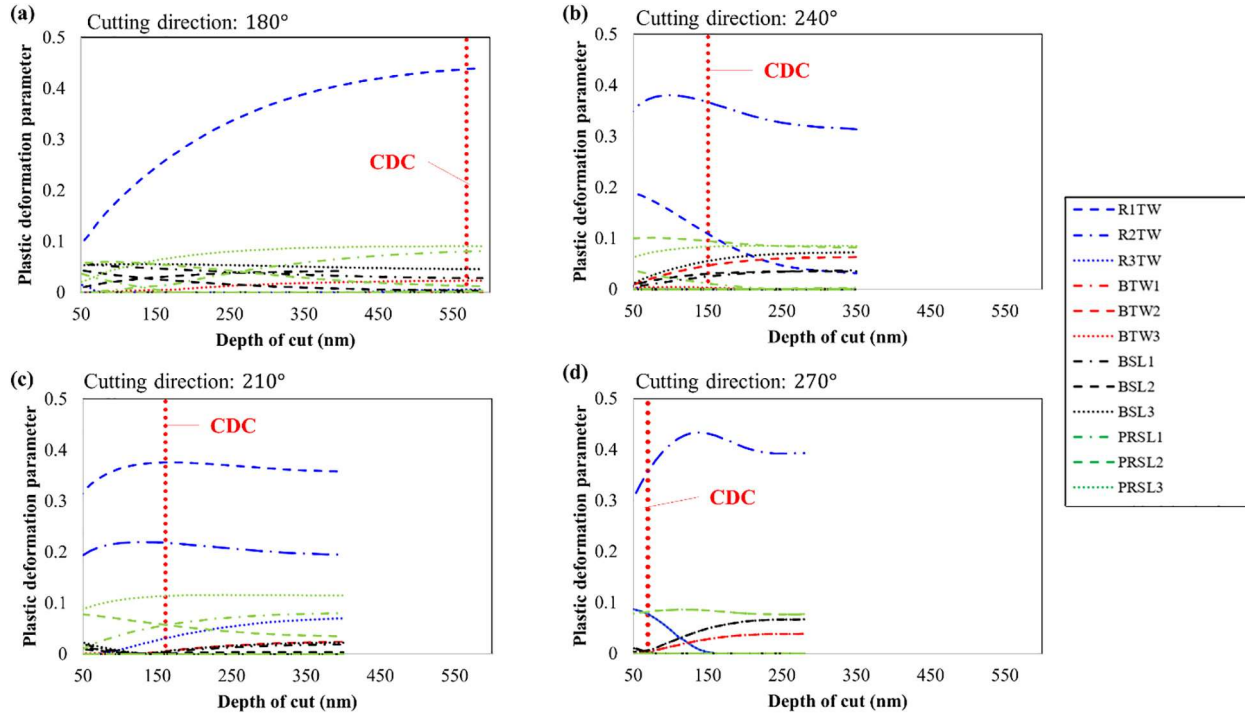


Fig. 21. Comparison of plastic deformation parameter and critical depth of cut in 210°, 240°, and 270° cutting directions with those in 180° cutting direction.

4.4. Empirical model for critical depth of cut

Fig. 22 shows that the resultant force and depth of cut are linearly proportional to each other and are related by the same constant of proportionality regardless of cutting direction when the depth of cut is smaller than the critical depth of cut. However, when the depth of cut exceeds the critical depth of cut, this relationship is no longer valid, and the resultant force fluctuates because of crack generation.

Fig. 23(a) shows the critical depth of cut and magnitude of measured resultant force at the transition point in terms of cutting directions. As shown in the figure, the profiles of resultant force and the critical cutting depth are almost the same in terms of cutting directions. This implies that larger amounts of plastic deformation results in a higher magnitude of resultant force. Fig. 23(b) shows the relationship between critical depth of cut and resultant force prior to the ductile-brittle transition, which can be expressed as follows.

$$F_{\text{crit}} = C \cdot \text{CDC} \quad (8)$$

where F_{crit} is the resultant force at ductile-brittle transition with units of N, C is the proportionality constant with value of 4.286×10^{-4} with unit of N/nm, and CDC is the critical depth of cut with unit of nm. The results show that the resultant force is proportional to the depth of cut until the critical depth of cut irrespective of the cutting direction. Once this value is reached, stress is relieved by crack initiation and the force values change based on cutting direction. Considering the unit of force can be expressed as $\text{kg} \cdot \text{m}/\text{s}^2$, unit of C can be expressed as kg/s^2 . The exact implications of the unit of C , kg/s^2 , and the physical meaning of the empirical equation needs further evaluation.

5. Conclusions

In this study, an updated model was developed to accurately describe plastic deformation and fracture during ultra-precision machining on R-plane of sapphire based on direction of resultant force. Fracture cleavage parameter and plastic deformation parameters

showing tendency of fracture/slip activation of each crystal plane were calculated and a detailed explanation of the crack morphologies and critical depth of cut was provided. Various crack morphologies can be predicted through geometric consideration of fracture parameters and interaction of activated systems. To summarize,

- For 90° cutting orientation, pull out crack was observed due to fracture system activation of both basal C- and prismatic M-planes which have opposite direction to each other.
- For 120° and 150° cutting orientations, triangular cracks were observed from a combination of horizontal crack by fracture activation of basal C-plane and diagonal crack by fracture activation of rhombohedral R-, prismatic M- and prismatic A-planes.
- For 180° cutting orientation, spalling cracks were generated by fracture activation of prismatic A-plane and radial crack was generated by fracture activation of prismatic M- and rhombohedral R-plane. Specifically, for spalling crack morphology, it is generated by coupling of fracture and slip activation of prismatic A-plane.
- In 210° to 270° cutting orientations, mainly spalling cracks could be observed by fracture activation of prismatic A-plane. For 210° cutting direction, M-shaped cracks could be also observed by fracture activation of prismatic A- and prismatic M-planes.

Through the analysis with plastic deformation parameter in terms of depth of cut until ductile-brittle transition, the anisotropy in the critical depth of cut can be explained by the dominant plastic deformation parameter. The modified slip/fracture model can be applied to other planes of sapphire or other single crystal materials in order to predict the crack morphologies and degree of critical depth of cut. In addition to this, it was found that the critical depth of cut is proportional to the resultant force as the amount of plastic shearing is closely related to applied force. The empirical relation is beneficial as it provides the foundation to control the ductile-brittle transition during machining. The ductile-brittle transition can be predicted by a controllable parameter and the resultant force. Further, from the fact that the empirical equation can be applied to the R-plane regardless of its anisotropy, one can

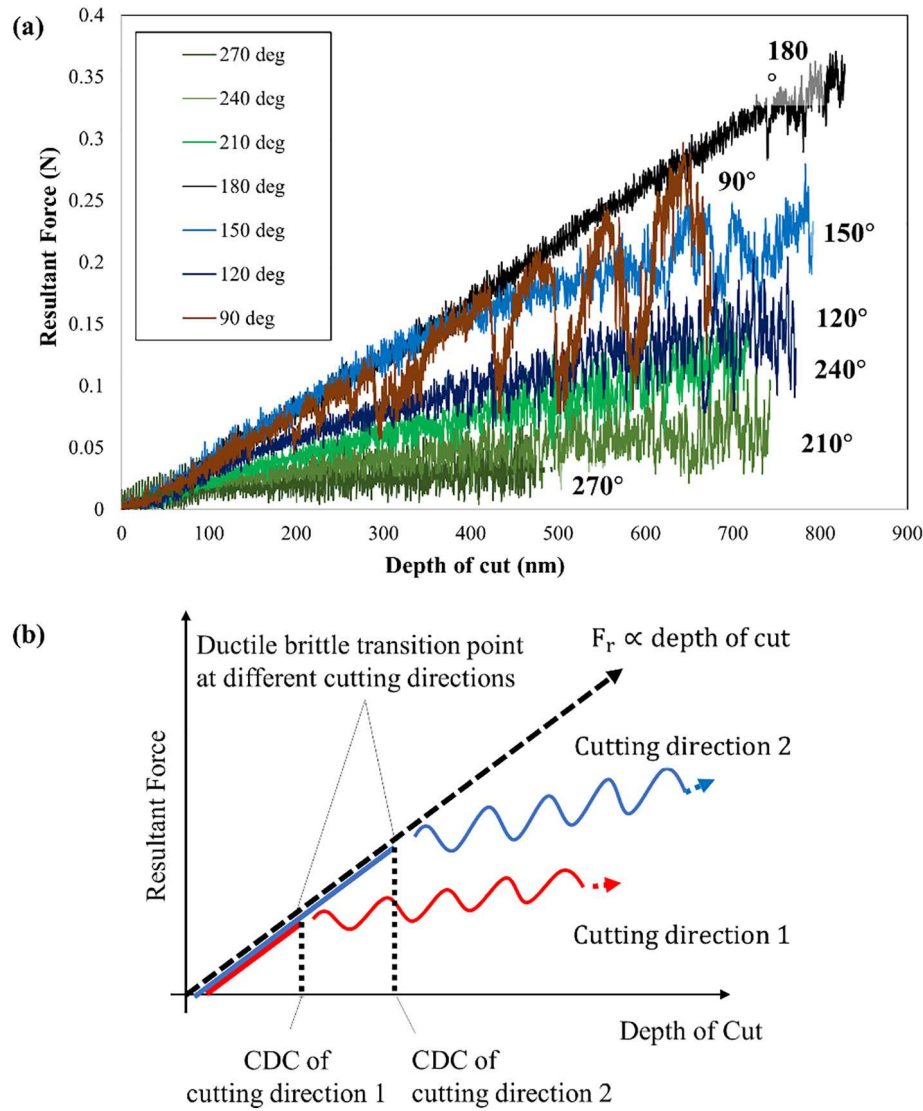


Fig. 22. Relationship between resultant force and depth of cut in terms of cutting direction at (a) before critical depth of cut and (b) after critical depth of cut.

expect that this equation can be applied to other planes as well. In future research, this model will be tested on other planes of sapphire and will be enhanced to describe how the material properties and machining input parameters influence the resultant force at the critical depth of

cut. In addition, further study will be conducted to understand how linear relationship (constant C in Eq. (8)) between resultant force and critical depth of cut varies for different sets of machining input parameters and its physical meaning.

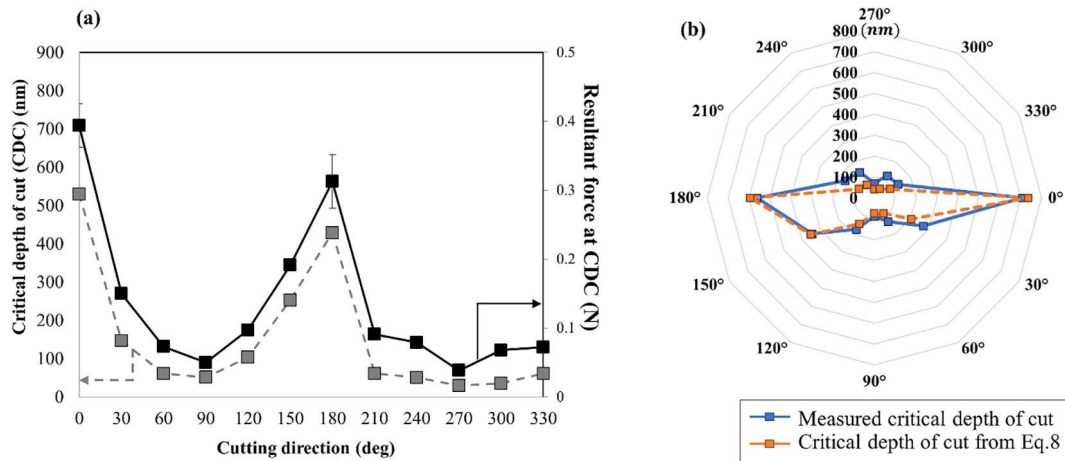


Fig. 23. (a) Critical depth of cut and resultant force in terms of cutting directions, and (b) critical depth of cut from measurement and Eq. (8), $F_{crit} = C \cdot CDC$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Suk Bum Kwon Ph.D. candidate in the Department of Mechanical Engineering, University of Wisconsin-Madison, Wisconsin, United States. His research focuses on modeling of ceramic material behavior during ultra-precision machining of ceramics.



Aditya Nagaraj is a Ph.D. candidate in the Department of Mechanical Engineering, University of Wisconsin-Madison, Wisconsin, United States. His research interests include ultra-precision machining of single crystalline ceramics, residual stresses and subsurface damage formation during machining.



Hae-Sung Yoon is currently an Assistant Professor in the School of Aerospace and Mechanical Engineering at Korea Aerospace University (KAU), Korea. He received his B.S. in 2010, and Ph.D. in 2015 (M.S. integrated) all in Mechanical and Aerospace Engineering from Seoul National University, Korea. Since then, he has held professional positions as a Postdoctoral Fellow in BK21+ Program at Seoul National University from 2015 to 2016, and as a Research Associate in the Department of Mechanical Engineering at University of Wisconsin-Madison, USA, from 2016 to 2018. Yoon's research interests are in Hybrid/Smart Manufacturing, Sustainable Manufacturing, Ultra-precision Machining, Micro/Nano Fabrication, 3D/4D Printing, Manufacturing for Design, and Smart Materials.



Sangkee Min is currently an assistant professor at Department of Mechanical Engineering, University of Wisconsin-Madison. He received Ph.D. in 2001 at U.C. Berkeley. He has been working on machining processes, burr formation, and smart manufacturing, and MFD (Manufacturing for Design).